SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

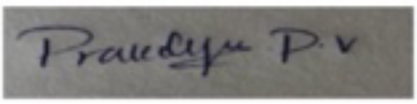
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I P V Prakalya (192311418) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

P.V.Prakalya CSA4305- Internet Programming 06/08/26

192311418 AT-03- CO2

Assignment Tool 3

1) Responsive Web Design

Responsive web design approach that masks a website automatically adjust its layout, images and current to fit differently screen sizes such as mobiles, tablet and desktop.

Importance of Responsive web design

→ works on all devices

→ Improves user experience

→ Faster and easier navigation

→ Reduces zooming and scrolling

→ Improves SEO

→ Serves development and maintainance cost

Key CSS Techniques for responsiveness:

1) Media Queries

→ Apply different CSS styles for different screen sizes

ex:-

```
@media (max-width : 768px) {
    .container {
        flex-direction: column;
```

}
}

Flexible Layouts

→ Use percentages (%) instead of fixed pixels

eg:- width : 80%

3). Flexible Images

→ images resize automatically

```
img {
  max_width : 100%
    height : auto;
  }
```

Using CSS Flexbox:

Ex: code

```
• container {
   display : flex ;
     gap : 10px
   flex -wrap : wrap;
 }
• box {
   flex : 1;
   Padding : 20px;
 }
```

Using CSS Grid :

Ex:- code

```
• container {
  display : grid;
grid -template -: repeat
  gap : 10px;
}
@media (max.width-768px)
  • container {
  grid - template -column : 1
  }
}
```

Simple Example Layout :

```
+-----------------------------+
|         Header              |
+-----------------------------+
| navigation menu             |
+-----------------------------+
| Content | sidebar           |
+-----------------------------+
|        footer               |
+-----------------------------+
```

```
+------------------+
|  Header          |
+------------------+
|  navigation      |
+------------------+
|  content         |
+------------------+
|  sidebar         |
+------------------+
|  footer          |
+------------------+
```

Improvements for better user Experience :-

→ Use responsive navigation

→ Increase button and textstyle on mobile

→ optimize images for better loading

→ Use readable font and colours

▸ client-side validation :

Client side validation is the process of checking user input in the browser using javascript before the submission of the form to the server.

purpose of client side validation;

→ prevents invalid data entry

→ improves user experience

→ reduces serverload

→ provide instance error messages

→ saves time by avoiding unessary from submission

JavaScript event handling in validation ;

- Javascript uses event to validate input fields

→ common Events :-

- onsubmit → check all fields before submitting
- on input → validates while typing
- onblur → validates when user leaves a field.

Coders :- [HTML]

```
<form onsubmit ="return validate Form ()">
Name : <input type=" text" id= "name">
<input type =" submit "value =" submit" >
</form>
<script>
notion validateform (){
let name = document. get ByIdd ("name") value;
  if (name =="" ){
    alert ("Name is required ");
  return False ; }
  return True ; }
  </script>
```

Role of regular Expressions (regex)

Regular Expressions are patterns used to check whether user input matches a format

Email Validation:

```
let
emailPattern = /^[A-Za-z0-9]._ /.+]+@[A-Za-z0-]
-/.[A,Z,a-z]{2,4$/;
```

Phone number validation:

Input

→ Name: Prakalya

→ Email: abc@gmail.com

→ phone: 9855477201

If Valid

→ Email: abcgmail com

→ phone : 96554

Output

Form submitted successfully

Improvements for better usability and security.

- Show clear error messages
- Highlight invalid fields
- Validate inputs in realtime
- Use placeholders eat input hints
- Disable submit button

CSS Box model and Responsiveness layout:

→ The css Box model defines how size of an HTML element is calculated.

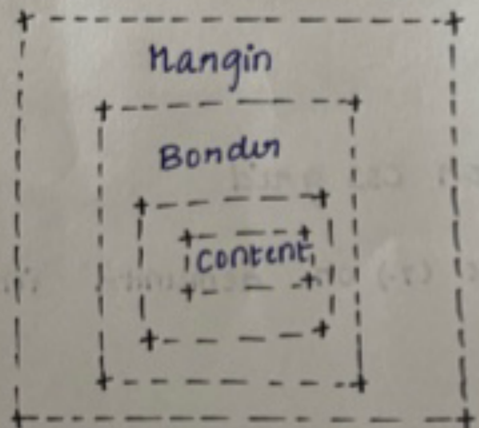
→ Every moment consists of 4 parts:

i) Content - Actual text on image

ii) Padding - space between content and border

iii) Border - surroundings padding

iv) margin - space outside the border



CSS positioning Techniques:

→ Static: Default position

→ Relative: moves relative to its original position

→ Absolute: positioned relative to nearest.

→ Sticky: Acts as relative until scrolling then becomes fixed

This affect how elements are arranged on webpages

Using media Queries

→ media Queries apply different css styles for different screen sizes

Ex:-

```
.container {
    display: flex;
}
@media (max-width : 768px) {
    .container {
        flex-direction: column;
    }
}
```

Best Practices

→ Use Flexbox or css grid

→ Use percentage (%) or remunits instead of fixed pixels

→ make image responsive using max-width : 100%.

→ Test the website on different screen sizes.

4). Document object Model (DOM) and Dynamic content:

DOM

→ The document object model is a tree-like structure created by the browser that represents HTML elements of a webpage

→ Javascript uses DOM DWOM to access, modify and add or to remove elements without reloading pages

Role of DOM in web development

- Access HTML elements
- Change text, images and styles
- Add or remove elements dynamically
- Handle user interactions
- update webpage content without refreshing.

Javascript DOM manipulation:

→ getElementById ()

→ query Selector ()

→ text content

→ style